Depth, Not Fidelity: What Decides Whether a Ternary Correction Helps a Mixture-of-Experts

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Abstract

Post-training ternary quantization (1.69 bits/weight) makes very large mixture-of-experts models runnable on consumer unified-memory hardware — we quantize a 397B-parameter, 512-expert MoE to 84 GiB and a 35B sibling to 9.2 GiB, both served by a modified llama.cpp with Vulkan ternary kernels. We then study a *corrective* mechanism: a second ternary plane on the hottest experts, which halves weight reconstruction error (46% $\to$ 22%) and halves the single-expert output error under real activations. **Applied uniformly, the correction makes both models worse.** We localise the damage with per-layer, per-projection switches and find that everything is decided by *where* the correction lands, not by *how faithful* it is: the identical perturbation, delivered at identical relative magnitude to every layer's output, improves the model in the final quarter of the depth and is catastrophic near the middle (a single layer moves perplexity from 8.72 to 37.8; the same correction at the last layers *improves* it). The fraction of depth at which the sign flips is the same — 73% — on both models, at a 10$\times$ size difference. We refute eight candidate explanations with pre-registered predictions — correction magnitude, routing concentration, expert-selection quality, input kurtosis, effective rank, permutation errors, delivered output error, and route-flip counts, the last of which *anti-correlates* with damage. The beneficial configuration is confirmed by a paired logit-level comparison (mean $\Delta p = +0.048\% \pm 0.014\%$), ships as a file rather than a runtime flag, and improves every evaluation chunk. At the task level, on an internal evaluation suite (aggregate results only; not publicly released), the shipped 1.75 bits/weight file resolves 74.7% of 146 paired coding-and-agentic trials against 55.5% for a size-matched IQ1_M quantization of the same model (exact McNemar $p = 2.5 \times 10^{-5}$) — matching the task success of a same-family 35B served at ~6.5 bits/weight. The same suite exposes a dissociation that perplexity inverts: the GPTQ-calibrated forge wins with reasoning off but degenerates into termination-token loops with reasoning on, while an RTN variant at the same bit-rate reasons correctly. Alongside, we report the first measured *distribution* of the MoE top-k routing margin: on 256 experts, the median gap between the 8th and 9th expert is 10$\times$ narrower than a uniform share, and 2.95% of tokens sit within one bf16 ULP of a routing flip — quantifying an assumption that stability lemmas and determinism patches currently take on faith.

1 Introduction

A 397-billion-parameter mixture-of-experts model does not fit in the memory of any consumer machine at the precisions its authors shipped. At 1.69 bits/weight it does: 84 GiB, resident in the unified memory of a desktop APU, served by ternary Vulkan kernels we contribute upstream. This paper is about what happened when we tried to make that model *better* — and measured, twice, that the intuitive way to do so makes it worse.

The corrective mechanism is a second ternary plane on the most-routed experts: a residual quantization that demonstrably halves weight reconstruction error (46% $\to$ 22%) and halves the single-expert output error under the layer's own captured activations. Every local metric of fidelity improves. Applied uniformly across the network, perplexity degrades — on both a 35B and a 397B model of the same family. This is the correction paradox, and it is not a bug: we verified the file layout, the expert permutations (28/28 correct on all 40 layers), the scales, and the delivered error, and then spent the balance of the work localising *where* the paradox lives.

The answer is depth. Per-layer engine switches (enabling the correction for arbitrary layer ranges at load time, without re-forging) show that the same perturbation, delivered at the same relative magnitude to every layer's output, improves the model in the final quarter of the network and is catastrophic near the middle: one single mid-depth layer moves perplexity from 8.72 to 37.8, while the correction restricted to the tail improves on the uncorrected baseline. The fraction of depth at which the sign flips is the same on both models — 73% — at a 10× parameter difference. Eight candidate explanations, each pre-registered as a falsifiable prediction before its measurement ran, are refuted in Section 4; what survives is a statement about position, not fidelity. Section 5 turns the statement into a recipe that ships inside the file. Section 7 reports a measurement the routing-stability literature has so far assumed: the distribution of the top-k routing margin, whose skew explains why the one published mean curve misleads.

We write from a deployment setting the quantization literature rarely occupies — one machine, no cluster, no fp16 reference for the largest model — and we are explicit throughout about what that setting does to measurement: paired estimators where absolute error bars would drown the effect, functional probes where benchmark harnesses disagree with each other, and a stated evaluation debt where neither suffices.

1.1 Contributions

1. **A depth rule, measured twice.** A partial ternary correction helps only in the final ∼quarter of the network, with the sign flipping at the same relative depth (73%) on two MoE models 10× apart in size. The recipe that follows — correction baked into the file for the tail only — improves the 35B by 5.4% and the 397B measurably (paired, 3.4σ), with no functional regression on verifiable-answer probes.

2. **Eight pre-registered refutations.** Every local account of the damage is excluded by measurement, including the decisive one: the perturbation *delivered to the layer's output*, $\|\Delta W \cdot X\|/\|WX\|$ under that layer's own captured activations, is constant across depth (1.000–1.002 ratio to the weight-space figure) and slightly *smaller* where damage is greatest ($\rho = -0.599$). Route-flip counts nearly invert the damage ordering (all-layers: 5× the flips of the worst single layer, 3.5× less damage).

3. **Task-level evidence at matched size, and a quantizer-level dissociation.** On an internal paired suite (two fixed seeds, 146 paired trials, identical server, reasoning off), the shipped file resolves 74.7% against 55.5% for a size-matched IQ1_M quantization of the same model (discordant pairs 36–8, exact McNemar $p = 2.5 \times 10^{-5}$; paired bootstrap CI_{95} of the difference $[+11.0, +27.4]$ points). With reasoning *on*, the ranking inverts: the GPTQ-calibrated planes degenerate into termination-token loops while an RTN forge at the same bit-rate passes every reasoning probe — a dissociation between quantization algorithms that neither perplexity nor no-think task rate detects (Section 6).

4. **The routing-margin distribution.** Defined twice in prior work (ReMoE's γ in probability space; a logit-space "router margin" reduced to one AUC) and never measured: we give percentiles. Median 10.1× narrower than the uniform expert share; 1st percentile 806× narrower; 2.95% of tokens within one bf16 ULP. The distribution is skewed (mean/median 1.76), which is precisely why the published token-averaged mean misleads.

5. **An engineering result.** TQ1_0 ternary kernels for Vulkan (upstream PR), a forge that produces two-plane ternary MoE files from bf16 on one consumer GPU (7.6 h for 397B), a strip tool that bakes measured layer profiles into files, and — new with this paper — a dense-model forge whose first target tests whether the depth rule needs experts at all.

1.2 What we explicitly do not claim

- Not "more bits hurt": the claim is *heterogeneous* fidelity hurts, and the literature contains the same phenomenon unremarked (GEMQ Table 1: expert-level mixed precision losing to uniform at matched bits (Deng et al., 2026); HiFloat4: restoring one component to high precision scoring below uniform FP4 (Mak et al., 2026)).

- Not the concept of the routing margin (ReMoE (Zhu et al., 2026) defines it), nor the down-projection bottleneck (D^2Quant (Yan et al., 2026) calls it "well-known"), nor non-additivity of compression error (EvoPress (Sieberling et al., 2025) opens on it). Our claims are the measurements, the localisation, and the rule.

- Not a mechanism. Eleven hypotheses died; the depth rule and the representational-transition literature (five methods placing a boundary at 0.47–0.53 of depth; our worst layer sits at 0.488) are consistent, but consistency is not identification. We state the open problem as narrowly as the measurements allow.

2 Setup and measurement honesty

Hardware. Everything runs on one consumer machine: Ryzen AI Max+ 395, 128 GB unified memory (∼96 GiB exposed as Vulkan VRAM), Fedora, our llama.cpp fork with TQ1_0 Vulkan kernels (scalar, coopmat1 and coopmat2 paths; the coopmat2 decode functions were later validated on an RTX 4060, the first hardware to exercise them). A second consumer laptop (RTX 4060 Laptop 8 GB) serves as an auxiliary bench for the 35B model.

Models and forge. The 397B (512 experts × 60 layers, hybrid GatedDeltaNet attention) is forged from the bf16 checkpoint: expert tensors ternarized with a two-level optimal-scale search plus true GPTQ (Frantar et al., 2023) against per-layer input Hessians (16,640 samples each); non-expert tensors from an existing shelf quantization. 7.6 h end-to-end. The 35B sibling is forged the same way from a shelf GGUF. Engine switches allow enabling the second plane per layer range and per projection at load time, which is what makes the per-layer anatomy measurable without re-forging.

What our perplexity numbers are, and are not. All perplexity is llama.cpp's, wikitext-2, ctx 2048, chunk counts stated per table. Absolute error bars at 30 chunks are ±0.087, so any comparison inside that band is resolved with the *paired* estimator (`-kl-divergence-base`): same chunks, reference logits stored, uncertainty propagated from the logits. The headline confirmation (tail-only correction on the 397B) is paired: mean $\Delta p = +0.048\% \pm 0.014\%$.

What our task numbers are, and are not. llama.cpp's built-in scorers label their output `acc_norm` but normalise endings by **token count**; lm-eval's `acc_norm` normalises by **character length**, and for Winogrande and MMLU lm-eval has no `acc_norm` at all. Our HellaSwag/Winogrande figures are therefore a third quantity, comparable only within this paper. At $n = 300$ their resolvable difference between two builds is 6.9 pp against typical quantization effects of 1–4 pp: we use them solely as "the model is not broken" evidence and never as a measurement of quantization damage. The final evaluation (pinned lm-eval via a server that implements prompt-token logprobs; `acc` and `acc_norm` both; full task sets) is listed in the debt section, and we explicitly do not use IRT-subsampled minibenchmarks: the published gp-IRT weights put only 12–30% of the reported figure on responses actually observed, calibrated on a 2024 pool of full-precision models that excludes the near-chance region a sub-2-bit model can occupy — a simulation on the published parameters shrinks true damage in the flattering direction.

The task suite. Because the standardised benchmarks above cannot resolve quantization-sized effects at the sample sizes we can afford, our task-level evidence comes from an internal private evaluation suite (91 tasks in total; suite not publicly released). We report the 73 tasks in the seven families and legacy set that are unrelated to the deployment environment — runtime bug-fixing, refactoring, test authoring (mutation-scored), long-context needle retrieval, tool use, deliberately deceptive code, planning, and a legacy mixed set; the remaining tasks are tied to the deploying household and withheld together with the suite itself. Aggregate results only: the suite supports the *paired* comparison between two files run under identical conditions, and we make no reproducibility claim on it — reproducibility is promised only on the standard benchmarks of the preceding paragraph. Each reported fixture is a self-contained repository with target tests and regression tests; a trial counts as resolved only if all target tests pass and no regression breaks (tool-call fixtures are scored on the emitted call transcript against a per-fixture contract). Every model runs every fixture at two fixed seeds (7 and 42), same llama.cpp server build, same 16,384-token context budget, reasoning off — 146 reported trials per model, paired trial-for-trial, so the comparison statistic is exact McNemar on the discordant pairs plus a paired bootstrap (20,000 resamples) for the interval.

Pre-registration discipline. Every hypothesis in Section 4 was written as a falsifiable prediction before its measurement ran, in the project log that ships with the artefact. Two hypotheses were "confirmed" by a first measurement and killed by a second (residual-stream rank; selection quality); both stages are kept in the record.

3 The paradox and the per-layer anatomy

The paradox, stated with its numbers. On the 35B, the second plane halves the weight reconstruction error of every expert it touches and halves the single-expert output error under captured activations. Enabled everywhere: perplexity 8.72 → 10.91. Every micro-level metric says the file got closer to the source model; the model says otherwise.

Coarse localisation. The per-projection switches split the damage immediately: `up`-only 8.42 and `gate`-only 8.52 are both *better* than baseline, `up+gate` better still (8.35) — and `down`-only alone is 11.24, worse than everything enabled together. One projection carries all of it, the same projection D^{2}Quant (Yan et al., 2026) calls a "well-known bottleneck" — except here the bottleneck is not low-bit difficulty but the *correction itself* doing harm.

Fine localisation. Layer-range switches on `down` walk the damage to its address: below the mountain (1–12) nearly harmless (9.02), above it (28–39) neutral-to-helpful (8.71, and 8.36 for 30–39), the middle catastrophic — 13–27 gives 25.34, and bisection lands on a single layer: **layer 20, alone, 8.72 → 37.80.** One layer of forty, corrected — by a mechanism that demonstrably improves that layer's own reconstruction — multiplies perplexity by 4.3. Neighbouring layers step down smoothly (25: 22.5; 30: 8.69), so this is a region, not a defect in one tensor; the boundary-check asserts and the permutation audit (Section 4, #6) close the trivial explanations.

The same anatomy at $10\times$**.** The 397B repeats the shape with its own geometry: the head is catastrophic, the middle harmful, and from layer 44 of 60 — 73% of depth, the same fraction as the 35B's 30 of 41 — the correction turns beneficial. The magnitude scales down with model size (the 397B's worst bands cost points, not multiples), which is itself consistent with the redundancy argument the MoE-robustness literature makes; the *sign structure* is what transfers exactly.

Why "paradox" is the right word. Sections 4–5 will show the perturbation delivered to each layer's output is depth-invariant (ratio 1.000–1.002 to the weight-space figure): the network is not receiving a bigger insult at layer 20 — it is receiving the *same* insult at a depth that cannot absorb it. Heterogeneous fidelity, not insufficient fidelity, is the failure mode.

4 Eight refutations, each with its pre-registered prediction

The method note that carries the section: every hypothesis was written down as a falsifiable prediction *before* its measurement ran, and two of the eight were "confirmed" by a first measurement and then killed by a second. We keep both stages in the record, because the difference between a correlation and a mechanism is precisely what this section is about.

Two cautionary subplots, reported rather than smoothed away:

#	Hypothesis	Pre-registered prediction	Measurement	Verdict
1	Correction magnitude	$\|\Delta W\|/\|W\|$ larger where damage is larger	0.412–0.423 at every layer (2% spread); $\rho = -0.56$	refuted
2	Routing concentration	hot-28 traffic share lower where damage is larger	$\rho = +0.013$; worst and best layers within 1.1 points of share	refuted
3	Scale mismatch (expert level)	cold-expert compensation helps wherever plane-2 is on	helps `down`-only (-0.14), harms all-three ($+0.70$)	refuted
4	Scale mismatch (layer level)	compensation helps a lot on the single worst layer, little on all layers	37.80 → 37.67 (0.5% of the damage); *harms* the uniform case	refuted
5	Input pathology (kurtosis / effective rank of the `down` input)	statistic peaks at the damaged layers	kurtosis $\rho = -0.19$; effective rank $\rho = +0.28$; layers 10 and 15 nearly identical rank, 2.3× different damage	refuted
6	Hot-first permutation error	wrong expert order at damaged layers	28/28 correct on all 40 layers, no exceptions	refuted
7	Delivered output error	$\|\Delta W \cdot X\|/\|WX\|$ larger where damage is larger	ratio to weight-space error 1.000–1.002 at every layer; $\rho = -0.599$	refuted — the decisive one
8	Expert-selection quality (frequency vs impact)	bands with near-perfect selection must benefit from the correction	397B band 30–45: overlap 25–27/28, best in the model — and it *harms* (6.3002 vs 6.1845)	refuted in causal form

Table 1: Eight refutations, each with its pre-registered prediction.

- **The residual-stream rank correlation that did not replicate.** On the 35B, effective rank of the residual stream correlates with damage at +0.651 — publishable-looking. On the 397B the sign *reverses* (damage highest where rank is lowest). One model would have made this a mechanism; the second model made it a coincidence. We keep it as the section's epigraph.
- **The selection hypothesis died twice.** Its correlational form is strong (impact lost by frequency-selection tracks the 397B damage profile at +0.788, and the head — 2/28 correct, 89.5% of impact lost — is exactly the catastrophic band). Its causal form failed the pre-registered test above. Selection quality is a real, quantified property of the forge and it does not decide the sign. Both statements are true; conflating them is how the literature acquires mechanisms it does not have.

What survives all eight is a statement about *position*: an identical perturbation, delivered identically, has consequences that depend only on the depth at which it lands — beneficial in the final quarter (from 73% of depth on both models), catastrophic near the middle of the 35B (its worst layer sits at 0.488 of depth, on the representational boundary that five independent methods place at 0.47–0.53), and progressively less harmful with depth on the 397B. Route-flip *counts* do not order the damage (they nearly invert it), so if routing mediates the effect it does so through *which* routes change, not how many — consistent with the one published attempt to classify flips, which achieves chance (AUC 0.490) at telling harmful from benign (Gu, 2026).

5 The depth rule and the recipe in the file

The rule. On both models, the second plane helps if and only if it is restricted to the final ∼quarter of the depth. On the 35B (41 layers, counted 0–40) the beneficial band is 30–39: 73.2% of depth. On the 397B (60 MoE layers with routed experts, band 44–59): 73.3%. Between the bands and the mountains sits no gradual transition on the 35B — layer 20 (0.488 of depth) alone accounts for most of the uniform-application damage — while the 397B decays monotonically from the head (catastrophic, 2/28 selection overlap) through the middle (harmful) to the tail (beneficial). The coincidence of the two flip points at 73% of depth, across a

Family	Task type	n	Ternary (ours)	IQ1_M
F1	runtime bug-fixing	18	12 (67%)	4 (22%)
F2	refactoring	18	12 (67%)	6 (33%)
F3	test authoring (mutation-scored)	18	8 (44%)	5 (28%)
F4	long-context needle	12	9 (75%)	11 (92%)
F5	tool use	12	12 (100%)	11 (92%)
F6	deceptive code	18	16 (89%)	11 (61%)
F7	planning	10	2 (20%)	2 (20%)
legacy	mixed original set	40	38 (95%)	31 (78%)
Total		146	109 (74.7%)	81 (55.5%)

Table 2: Paired results on the reported portion of the internal suite: 73 fixtures × 2 seeds (7, 42), reasoning off, identical server. Discordant pairs 36–8 in the ternary file's favour; exact McNemar $p = 2.54 \times 10^{-5}$; paired bootstrap (20,000 resamples) CI_{95} of the difference $[+11.0, +27.4]$ percentage points.

10× size difference and two different routing topologies, is the paper's central empirical fact. We claim the measurement, not a law: two models, one family (Section 9).

The recipe. Because the engine switches that made the anatomy measurable are ours, we could ship the rule as a runtime flag — and explicitly do not. A flag is a claim about every future file; a file is a claim about itself. The forge takes `-plane2-layers LO-HI` and bakes the tail-only correction in: plane-2 tensors exist only for the beneficial band, the expert permutation and router reordering are applied only where the plane exists (one predicate feeds all three coupled decisions), and a strip tool retrofits the same profile onto already-forged files, reclaiming the dead plane-2 bytes (4.1 GiB on the 397B). The resulting file runs on the unmodified engine at full speed with no configuration.

Confirmation, 35B. Tail-only (30–39) perplexity 8.2501 against 8.72 uncorrected — a 5.4% improvement — where uniform application had *degraded* the same model. Functional probes (verifiable-answer arithmetic and string-manipulation set) unchanged or improved; no regression observed.

Confirmation, 397B, paired. At 397B the absolute error bar at 30 chunks (±0.087) is larger than the expected effect, so the confirmation is paired (`-kl-divergence-base`, same chunks, reference logits stored): mean $\Delta p = +0.048\% \pm 0.014\%$ (3.4σ) for tail-only over uncorrected, improvement in every evaluation chunk, and the verifiable-answer probes intact. The gain is small and real; the point is its *sign*, which uniform application gets wrong at 10× the magnitude.

Cost. The tail-only file is strictly smaller than the uniform one (the correction exists on 16/60 rather than 40/60 layer-equivalents), loads nothing extra at the head where it would do harm, and the depth rule transfers across the two models tested at zero search cost — against EvoPress-style whole-model search (Sieberling et al., 2025), which subsumes the selection problem but must re-run per model.

6 Task-level evaluation at matched size

The comparison. The natural size-matched baseline for the shipped file is llama.cpp's own IQ1_M quantization of the same 397B checkpoint — the same 1.75 bits/weight on the expert tensors, produced by the standard importance-matrix pipeline rather than our two-plane ternary forge. Both files ran the suite of Section 2 under identical conditions (same server build, same seeds, reasoning off). Table 2 gives the per-family results for the reported portion.

Reading the table honestly. The aggregate difference (+19.2 points) is carried by the code-understanding families — bug-fixing, refactoring, deceptive code — where the ternary file wins by 22–45 points. The one family the baseline wins as scored, long-context needle (75% vs 92%), we report as scored: post-hoc inspection attributes all three ternary misses to infrastructure failures (context-window overflow and timeouts), and the ternary file is 9/9 on the trials that ran to completion, but the scored number stands. Planning (F7) is the shared floor: 20% for both giants with reasoning off, discussed below and in Section 9. For context only — an earlier version of the internal suite, so not a controlled comparison — the strongest previously deployed configuration on this machine, a same-family 35B served at ∼6.5 bits/weight, scores 75%: the 397B at 1.75 bits/weight reaches the task-success level of a full-working-precision 35B on the same hardware budget.

6.1 A quantizer-level dissociation: reasoning survives RTN, not GPTQ

The result we did not expect from this suite concerns the quantization *algorithm* rather than the file. Our forge has produced, at the same bit-rate and the same two-plane layout, first-plane variants quantized by plain optimal-scale RTN and by GPTQ against generic-text calibration Hessians. On every proxy we had trusted so far, GPTQ wins: better perplexity, and the no-think task rates of Table 2. With reasoning *enabled*, the ranking inverts. The RTN build produces sane chains of thought and passes all three hard verifiable probes (multi-step percentage arithmetic; a classic kinship-counting trap; character-level string reversal). The GPTQ builds, on the same probes, degenerate into unterminated loops of the thinking-close token and never converge to an answer — a failure invisible to perplexity and to no-think task rate alike.

This dissociation is consistent with the emerging picture of calibration-distribution mismatch in low-bit reasoning models. Quantization is known to damage reasoning disproportionately at low bit-rates (Liu et al., 2025), with looping and non-termination documented as characteristic failure modes of extreme low-bit inference (Alimaskina et al., 2026), and quantized reasoning models systematically inflating their chain-of-thought length without gaining accuracy (Lotfi et al., 2026). The remedy reported in that literature is distributional: calibrating on reasoning-style data rather than generic text recovers large fractions of the loss (Liu et al., 2025), and calibrating on the model's *own* generated reasoning traces recovers up to 25.5 points at 1.58 bits (Wang et al., 2026); practitioner recipes for 1.58-bit reasoning models likewise survive only by protecting the components that generic pipelines damage (Han & Han, 2025). Our RTN/GPTQ pair isolates the same variable within a single forge: GPTQ minimises error *on the calibration distribution* and displaces it off-distribution — and long self-generated reasoning traces are precisely off-distribution with respect to a generic text corpus, while RTN's error is distribution-agnostic. We state this as the consistent explanation, not a demonstrated mechanism; the discriminating experiment (re-forging the GPTQ planes against self-generated reasoning traces, ScaleQ-style) is scheduled and its outcome is falsifiable in the same way as Section 4's hypotheses. Until it lands, the shipped file runs with reasoning disabled in production, and Table 2 is a no-think table by construction.

7 The routing-margin distribution

Definition. For a token routed to k of E experts (Fedus et al., 2022), the margin is the gap between the k-th and $(k+1)$-th router score — the distance to the nearest routing flip. ReMoE (Zhu et al., 2026) defines it in probability space as the premise of a stability lemma; Gu (2026) defines it in logit space and collapses it to a single AUC. Neither reports its distribution; Chen et al. (2026) reports a token-averaged mean over depth. We measure the distribution on the 397B's captured router probabilities (16,640 tokens × 60 layers, the same capture that feeds the forge's Hessians).

The distribution. The median margin is 10.1× narrower than the uniform share $1/E$; the 1st percentile is 806× narrower; 2.95% of tokens sit within one bf16 ULP of the 8th/9th boundary — for these, the routing decision is below the numerical noise floor of the arithmetic that computes it. The distribution is right-skewed, mean/median 1.76: the mean curve published in prior work sits 76% above the typical token, which is precisely the regime where a mean reassures and a median warns.

What follows and what does not. The qualitative half of this measurement was asserted, on synthetic weights, by the author of sglang PR #35916 (pol-resina, 2026) — batch-invariant routing motivated by adjacent scores within one ULP on this very topology — who lacked hardware for the real model; this section supplies the measurement. What does *not* follow is the tempting link to Section 4's damage profile:

route-flip counts anti-correlate with damage (the all-layers configuration flips 5× more routes than the worst single layer at 3.5× less damage), and the one published flip classifier sits at chance (AUC 0.490). Fragile margins are a fact of the router; they are not, by count alone, the mechanism of the correction paradox.

8 Related work

Ternary post-training quantization. PTQTP (Xiao et al., 2025) introduces the dual trit-plane decomposition we build on, applied uniformly, with no per-layer analysis. Tied Trit-Planes (Grella, 2026) is the closest work: two ternary planes on a disk-streamed MoE, same primitive, same deployment constraint — and it reports, without explanation, the same proxy/fidelity dissociation we measure (its higher-reconstruction-error variant scores *better* on MMLU). Its ladder is over tensor classes and, by its own statement, "localizes sensitivity to the final cumulative bundle, not to a single component"; per-layer localisation is precisely what we add. BitNet (Ma et al., 2024) trains ternary from scratch and keeps embeddings high-precision; our setting is post-hoc, on weights never meant for ternary.

Non-additivity of compression error. EvoPress (Sieberling et al., 2025) opens on the observation that per-layer independence fails and that removing *more* blocks can help; its search machinery (fitness = whole-model KL) subsumes our selection procedure. What it explicitly does not do is isolate single catastrophic layers or offer a mechanism. QDrop (Wei et al., 2022) and Quant-Noise (Fan et al., 2021) establish the non-monotone shape at calibration time — full quantization-aware noise *worse* than none, partial better than both — which is the same shape as our cancellation, in a different regime.

Mixed precision within MoE. The expert-wise bit-allocation literature (MoPEQ (Chitty-Venkata et al., 2025), AlphaQ (Yang et al., 2026), MC-MoE (Huang et al., 2024), GEMQ (Deng et al., 2026), BitsMoE (Zhao et al., 2026)) is uniformly monotone in its assumptions: the only failure mode contemplated is mis-estimating importance. The phenomenon we isolate is visible, unremarked, in GEMQ's own Table 1 (expert-level mixed precision losing to uniform at matched bits on Qwen3-30B-A3B, −9.89 acc at 2.0 bpe) and named in HiFloat4 (Mak et al., 2026): restoring one component to higher precision scores *below* uniform FP4. AWQ's Table 1 (Lin et al., 2024) shows the selection-criterion analogue: protecting the wrong 1% is worse than protecting nothing. Our framing — heterogeneous fidelity, not insufficient fidelity — is the synthesis these scattered rows point to.

Routing fragility. The margin at the top-k boundary is defined twice and measured never: ReMoE (Zhu et al., 2026, App. C.2) as the unmeasured premise of a stability lemma, in probability space; Gu (2026) in logit space, collapsed to one AUC (and its margin→harmful classifier sits at chance, 0.490 — consistent with our flip-count anti-correlation). A token-averaged mean curve exists (Chen et al., 2026); the distribution does not, and the mean-vs-median gap (1.76×) is exactly why it matters. sglang PR #35916 (pol-resina, 2026) asserts the qualitative half — adjacent scores within one bf16 ULP on this very topology — on synthetic weights, its author lacking hardware for the real measurement; we supply it.

Depth structure. The representational-transition literature places a boundary at 0.47–0.53 of depth by five independent methods (semantic hub, activation patching — concept written at exactly 0.500 — probing centers of gravity, and their encoder controls); our worst layer sits at 0.488. The intervention literature, by contrast, is dominated by edge-sensitivity: GLU activation spikes at early *and late* layers (Yang et al., 2024), llama.cpp's own extra-bit heuristic on the first and last eighths, and layer-pruning work finding deep layers most redundant (Men et al., 2024; Gromov et al., 2024). The tension is real and stated: those works measure activation magnitude or deletion tolerance; we measure *corruption* tolerance, and deletion-vs-corruption is exactly the distinction the reconciliation requires (a deleted layer passes the residual through; a corrupted one writes an actively wrong signal where the representation is being formed). One prior intervention study reports a mid-depth sensitivity peak (Xu et al., 2025, activation injection, mean peak 54.9%).

Down-projection difficulty. Known and named: D^2Quant (Yan et al., 2026) calls it "a well-known quantization bottleneck" and engineers a dedicated dual-scale quantizer. Community evidence on our very model family: unsloth's v3 recipe placing 20 large FFN tensors at 2 bits measurably degraded coding. What is not in the literature is the sign flip — `down` catastrophic on the mountain, *helpful* in the tail — nor any derivation from gated-activation statistics (our attempt at one is refuted by measurement: kurtosis, effective rank and delivered error all fail to track the damage).

9 Limitations

- **Two models, one family.** Both are Qwen-lineage MoEs (35B and 397B) with shared tokenizer and expert topology conventions. The 73% coincidence is measured twice, not derived; a third family is the obvious next falsification, and the dense-forge experiment (running as this draft is written) asks whether the rule needs experts at all — early evidence says dense FFNs at this bit-rate fail globally before any depth structure can be read, consistent with the MoE-robustness literature (Kim et al., 2023), which would make the depth rule a property of *corrections to survivable quantizations* rather than of quantization per se.
- **Perplexity-first evaluation.** Our confirmations are paired logit-level comparisons plus verifiable-answer functional probes; the standardised benchmarks we can run at $n = 300$ resolve 6.9 pp at best and are used only as "not broken" evidence. The pinned lm-eval pass is stated as debt, not waved at.
- **The selection confound is quantified, not closed.** Frequency-selection wastes 42.6% of plane-2 impact in the head and its correlational profile tracks damage at +0.788; the causal test refutes it as *the* mechanism, but a Hessian-impact-selected reforge (the artefact exists; the run is scheduled) is required before selection can be fully dismissed as a contributor to the magnitude, as opposed to the sign.
- **No mechanism.** We offer a rule and eight dead hypotheses. The representational-transition literature is consistent with the boundary we measure and predicts none of its consequences quantitatively. We consider naming this openly a feature of the paper, not a gap in it.
- **Planning is a shared floor, and reasoning is off.** The planning family of the task suite sits at 20% for *both* 1.75-bit giants with reasoning disabled — the suite comparison says nothing about which file plans better, only that neither does at this bit-rate without chain-of-thought. And reasoning is disabled in the shipped configuration precisely because of the GPTQ dissociation of Section 6.1: until the reasoning-trace recalibration closes that gap, the task-level claims of Table 2 are no-think claims and must not be read as more.
- **One quantization primitive.** Everything is TQ1_0-shaped ternary with two-level scale search and GPTQ; whether the depth rule survives a change of primitive (say, trellis-coded or lattice quantization at matched bpw) is untested. A preliminary result on real weights says the change of primitive is worth forging: an exact tail-biting Viterbi encoder over a 4096-state bitshift trellis (56 bytes per 256-weight block = 1.75 bpw exact), run on `blk.29.attn_gate` of the 397B (dequantized from the Q8_0 reference; 4096 blocks sampled uniformly at seed 0 — a declared subset on which the RTN baseline matches its full-tensor value to four decimals), reaches relative error 0.3473 at 1.75 bpw against 0.4420 for optimal-scale ternary RTN at 1.69 bpw: −38.3% MSE at near-matched rate. The measurement behaves like its theory: the gaussian simulation (0.3329) transfers to the real tensor with the same $\sim$1% excess that RTN shows over its Lloyd bound (0.4362 $\rightarrow$ 0.4420); tail-biting converges on 4093/4096 blocks in at most three extra passes; the packed 56-byte payload decodes bit-exactly to the same error; and the compute-based (1MAD) codebook costs $\sim$3% MSE that an 8 KiB lookup table would recover. This is a pre-test on one tensor, not a forged model: whether the tensor-level −38% survives GPTQ, a full forge, and — the question this paper makes precise — the depth rule, is exactly what remains untested.

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